

Purvesh Mandawkar

Embedded Systems Developer · IoT · PCB Design

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EDUCATION

B.E. Electronics & Telecommunication Engineering

2024 – 2028

Sinhgad College of Engineering, Pune — Currently 2nd Year

PROJECTS

Fail-Safe 220V AC Pump Controller

Industrial-grade control system for a 1HP water pump. Handles inrush surge (RC snubber), water turbulence (5s debounce filter), and MCU freeze (hardware watchdog). Manual bypass for zero-downtime operation.

[Power Electronics](#) · [Real-time Firmware](#) · [Watchdog Timer](#) · [Relay Control](#) · [Fail-safe Design](#)

IoT ECG & Heart Rate Monitor

Captures microvolt-level bio-signals via AD8232 analog front-end, performs R-peak detection on device, and streams real-time BPM to mobile dashboard over Wi-Fi using Blynk IoT platform.

[Analog Signal Chain](#) · [Biopotential Sensing](#) · [IoT Platform](#) · [Edge Processing](#)

Automated Soil Irrigation System

Closed-loop irrigation using capacitive soil moisture sensing and hysteresis control logic to prevent relay chatter — calibrated thresholds for water efficiency and crop health.

[Capacitive Sensing](#) · [Hysteresis Control](#) · [Relay Actuation](#) · [Embedded Firmware](#)

Autonomous Thermal Monitor & Alert System

Standalone battery-powered safety device on bare-metal ATmega firmware. Three-tier cumulative alert architecture with star-grounding and decoupling capacitors for noise-free standalone operation.

[Bare-metal Firmware](#) · [ATmega8A](#) · [Battery Optimization](#) · [Standalone Embedded](#)

AI Voice Chatbot — Edge Hardware [In Progress]

Alexa-like voice assistant on IoT edge hardware. Full voice pipeline from microphone capture to on-device inference, no cloud dependency.

[Edge AI](#) · [Voice Pipeline](#) · [Memory Optimization](#) · [IoT Development Board](#)

SKILLS

Embedded Development: IoT Development Boards · Bare-metal Firmware · Real-time Control · State Machines

Firmware & Programming: Embedded C · C++ (Arduino Framework) · Interrupt Handling · Watchdog Timers

PCB & Circuit Design: Schematic Design · PCB Layout · KiCad · Soldering

Hardware Systems: Analog Signal Chain · Power Electronics · Sensor Integration · Relay & Actuator Control

EVENTS & EXPOSURE

EFY Expo Pune 2025 Electronics and embedded systems expo — components, IoT hardware, product development trends.

Odoo Event 2025 Product and innovation-focused event — industry talks, demos, and scalable tech solutions.